



A MODEL OF DESTINATION IMAGE FORMATION

Seyhmus Baloglu

University of Nevada--Las Vegas, USA

Ken W. McCleary

Virginia Polytechnic Institute and State University, USA

Abstract: Image has been shown to be an important influence in the selection of vacation destinations. A model that represents the important determinants of destination image formation was developed based on previous studies in a number of fields. The research reported in this article presents the results of an empirical test of the model using path analysis. A major finding of the study was that a destination image is formed by both stimulus factors and tourists' characteristics. The results of this investigation provide important implications for strategic image management and can aid in designing and implementing marketing programs for creating and enhancing tourism destination images.

Keywords: image model, destinations, image formation, path analysis. © 1999 Elsevier Science Ltd. All rights reserved.

Résumé: Un modèle pour la formulation de l'image de marque des destinations. On a démontré que l'image de marque est une influence importante dans le choix de destination de vacances. En se basant sur des études précédentes dans plusieurs domaines, on a développé un modèle qui représente les déterminants importants de l'image d'une destination. Cet article présente les résultats d'un test empirique de ce modèle en utilisant l'analyse de voie. Une conclusion majeure est que l'image d'une destination est formée des facteurs de stimulation et des caractéristiques des touristes. Les résultats de cette recherche fournissent des implications importantes pour la gestion de l'image stratégique et peuvent contribuer à la conception et à la réalisation des programmes de marketing pour créer et mettre en valeur les images des destinations touristiques. © 1999 Elsevier Science Ltd. All rights reserved.

INTRODUCTION

Research of the past two decades has demonstrated that image is a valuable concept in understanding the destination selection process of tourists. Several studies centered on the relationship between destination image and preference or visitation intentions (Goodrich 1978; Mayo 1973; Hunt 1975; Milman and Pizam 1995; Scott, Schewe and Frederick 1978). A particular research stream investigated the impact of previous visitation (actual behavior) or familiarity on destination image (Ahmed 1991; Chon 1990; Dann

Seyhmus Baloglu is Assistant Professor in the Department of Tourism and Convention Administration at the University of Nevada (Las Vegas, NV 89154-6023, USA. Email <baloglu@cmail.nevada.edu>). His research interests in marketing focus on image, positioning, segmentation, and loyalty. **Ken McCleary** is Professor of hospitality and tourism marketing. His Ph.D. in marketing is from Michigan State University.

1996; Fakeye and Crompton 1991; Fridgen 1987; Hu and Ritchie 1993; Milman and Pizam 1995; Pearce 1982; Phelps 1986). Some studies examined the relationship between tourists' geographical location (distance) and image (Ahmed 1991; Crompton 1979a; Fakeye and Crompton 1991; Hunt 1975; Scott et al 1978). Others focused on the measurement of destination image (Echtner and Ritchie 1993; Driscoll, Lawson and Niven 1994), its components (Dann 1996; MacKay and Fesenmaier 1997), or factors influencing it (Baloglu and Brinberg 1997; Walmsley and Jenkins 1993). Still others examined temporal influences on image change (Gartner 1986; Gartner and Hunt 1987), differences between tourist image (demand) and what is projected by destinations (Stabler 1990), variations by trip purpose (Javalgi, Thomas and Rao 1992), and the relationship between sociodemographic variables and destination image (Baloglu 1997; Walmsley and Jenkins 1993). However, little empirical research has focused on how image is actually formed, especially in the absence of previous experience with a destination. This suggests that most studies have largely focused on its static structure by examining the relationship between image and behavior, but not on its dynamic nature by investigating the influences on its structure and formation in the absence of actual visitation.

The initial image formation stage before the trip is the most important phase in tourists' destination selection processes (Gunn 1972; Mercer 1971). As noted by Brokaw "Before image can be used to influence behavior, it is important to understand what influences image" (1990:32). Goodall (1990) noted that knowing factors influencing it would help identify target markets and decide which image should be promoted to which segment of the market. Numerous researchers across fields emphasized the importance of understanding forces which influence image development and suggested that little effort has been undertaken to determine the factors that influence its formation (Brokaw 1990; Burgess 1978; Fakeye and Crompton 1991; Gartner 1989, 1993; Han 1989; Kotler, Haider and Rein 1993; Myers 1968; Phelps 1986; Russel and Snodgrass 1987; Stern and Krakover 1993). Despite importance and growing interest, destination image studies have also been criticized as atheoretical and lacking a conceptual framework (Echtner and Ritchie 1993; Fakeye and Crompton 1991; Gartner 1993).

To address the problem of identifying what influences image development, a model of its formation was developed based on the literature from several fields and disciplines. The model is designed to provide a framework for studying the forces guiding the formation of destination image and proposes relationships among the different levels of evaluations within its structure (cognitive, affective, and global), as well as the elements determining these evaluations. The purpose of the research reported here was to test the proposed model and to provide insights into how images of destinations are developed.

DESTINATION IMAGE FORMATION

Figure 1 presents a general framework of destination image formation developed from previous literature. Numerous researchers across fields and disciplines agree that image is mainly caused or formed by two major forces: stimulus factors and personal factors. The former are those that stem from the external stimulus and physical object as well as previous experience. Personal factors, on the other hand, are the characteristics (social and psychological) of the perceiver. The general framework presented in Figure 1 served as a framework in developing the proposed path model in Figure 2 for this study.

The literature review of the influences on destination image revealed three major determinants existing in the absence of actual visitation or previous experience: tourism motivations, sociodemographics, and various information sources. In this regard, the latter represent stimulus variables whereas motivations and sociodemographics stand for consumer characteristics in the proposed path model. Each component was selected based on the literature review and research of other models. The image concept has generally been considered as an attitudinal construct consisting of an individual's mental representation of knowledge (beliefs), feelings, and global impression about an object or destination. Researchers in several disciplines and fields agree that the image construct has both perceptual/cognitive and affective evaluations. The perceptual/cognitive evaluations refer to the beliefs or knowledge about a destination's attributes whereas affective evaluation refers to feelings toward, or attachment to it. A common agreement is that this depends on a cognitive evaluation of objects and the affective responses are formed as a function of the cognitive responses. An overall image of a place is formed as a result of both perceptual/cognitive and affective evaluations of that place.

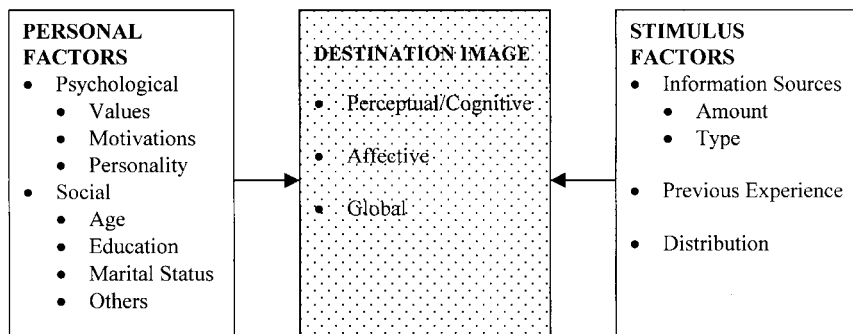


Figure 1. A General Framework of Destination Image Formation

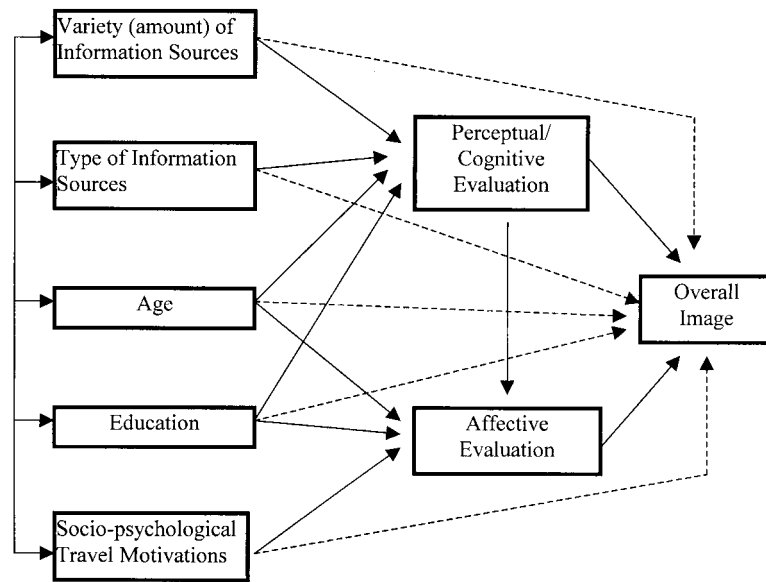


Figure 2. Path Model of the Determinants of Tourism Destination Image Before Actual Visitation. Note: Straight Lines from Exogenous to Endogenous Variables Denote the Paths (Effects) that are Hypothesized and Tested, While Dashed Lines Denote the Paths (Effects) that are not Hypothesized but Tested to Examine the Overall Pattern of the Model.

Components of the Path Model

Image. Academic interest in several fields and disciplines regarding the concept of image has been pervasive since the early works of Boulding (1956) and Martineau (1958) who proposed that human behavior is dependent upon image rather than objective reality. These early works and subsequent embracement of the image concept have led to "image theory" which suggests that the world is a psychological or distorted representation of objective reality residing and existing in the mind of the individual (Myers 1968). A commonly adopted definition of image is that it is a set of beliefs, ideas, and impressions that people have of a place or destination (Crompton 1979a; Kotler et al 1993). It is a mental representation of an object or place which is not physically before the observer (Fridgen 1987). Assael (1984) defined image as a total perception of a product that is formed by processing information from various sources over time. These definitions mostly emphasized perceptual/cognitive component of image.

Lawson and Baud-Bovy (1977) defined a destination image as the expression of all knowledge, impressions, prejudices and emotional thoughts an individual or group has of a particular object or place. Similarly, Oxenfeldt (1974–75) and Dichter (1985) viewed image as an overall or total impression which is formed as a result of the evaluation of individual attributes which may contain both cognitive

and emotional content. With this definition, Oxenfeldt (1974–75) and Dichter (1985) recognized not only cognitive and affective images, but also the formation of overall image from evaluations of an object. This view is supported by Mazursky and Jacoby (1986) who defined image as a set of cognitions and affects that represent an entity to an individual. A succinct review of brand image by Dobni and Zinkhan (1990) revealed that it is to a great extent a perceptual phenomenon which is formed through consumers' reasoned and emotional interpretation. They also concluded that an image is affected by both stimulus elements of the product and the characteristics of the perceiver.

Perceptual/Cognitive and Affective Components. Perceptual or cognitive evaluation refers to beliefs and knowledge about an object whereas affective refers to feelings about it (Baloglu and Brinberg 1997; Burgess 1978; Gartner 1993; Holbrook 1978; Walmsley and Jenkins 1993; Ward and Russel 1981; Zimmer and Golden 1988). People develop both cognitive and affective responses and attachments to environments and places (Proshonsky, Fabian and Kaminoff 1983). Holbrook (1978) noted that in marketing and consumer behavior those two components are treated under the label of "beliefs" vs. "affect."

Many findings in environmental psychology also support the notion that settings have both perceptual/cognitive and affective images (Burgess 1978; Hanyu 1993; Lynch 1960; Russel and Pratt 1980; Russel, Lewicka and Niit 1989). The perceptual/cognitive component is the knowledge about the place's objective attributes whereas the affective counterpart is knowledge about its affective quality (Genereux, Ward and Russel 1983). Hanyu suggested that "Affective meaning refers to the appraisal of the affective quality of environments while perceptual/cognitive quality refers to the appraisal of physical features of environments" (1993:161). The author also noted that environmental studies so far have either focused on perceptual/cognitive or affective meaning and no study has measured the two constructs at the same time. Some recent works, however, studied both cognition and affect toward environments and destinations (Baloglu 1998; Dann 1996; MacKay and Fesenmaier 1997).

Tuan (1975) noted that an image of environment or place is something formed in the human mind in absence of physical environment. After an extensive literature review of image in geography, Burgess (1978) concluded that most studies emphasized the physical qualities of places where, there is indeed an interaction between physical qualities of places and the meanings (affective ties) they acquire. Lynch (1960), in his seminal book *The Image of the City*, suggested that meaning or the emotional component of environmental image can be separated from its physical form and the emotional component is dependent on such attributes. Russel and Pratt (1980) separated affective from cognitive images and suggested that the link between latter (place attributes) and former

reactions should be studied to understand what attributes correspond to a given affective reaction.

A common agreement among diverse researchers is that affective evaluation depends on cognitive assessment of objects and the affective responses are formed as a function of the cognitive ones (Anand, Holbrook and Stephens 1988; Burgess 1978; Gartner 1993; Lynch 1960; Holbrook 1978; Reibstein, Lovelock and Dobson 1980; Russel and Pratt 1980; Stern and Krakover 1993). This suggests that although a distinction is made between the two dimensions, they are also interrelated. The distinction and direction of relationship between perceptual/cognitive and affective components has been emphasized in various consumer and tourism decision-making models (Crompton and Ankomah 1993; Mayo and Jarvis 1981; Woodside and Lysonski 1989). Mayo and Jarvis (1981:190) conceptualized a model of the tourism decision-making process with special emphasis on attitudes or images toward destinations. In this model, tourists form their feelings as a function of beliefs and opinions. When relating image formation to the destination selection process, Gartner (1993) proposed that its cognitive component (defined as the sum of beliefs and knowledge of attributes of the object or product), and its affective component are distinct but hierarchically related.

Overall Image. Both perceptual/cognitive and affective evaluations form the overall image of a place. Stern and Krakover (1993), in their model of the formation of a composite urban (city) image, depicted that designative (perceptual/cognitive) and appraisive (affective images) together form a composite or overall image of a city. Their results provided support for the intervening role of affect between perceptual /cognitive evaluation and overall image, as well as the interactive effects of the two components in forming overall image. Mazursky and Jacoby's (1986) model of store image formation described that after consumers evaluate and integrate perceptions of store attributes, they ultimately form an overall image which is the end-product of this formation process.

Gartner (1986:635–636) stated that people's perceptions of various attributes within a destination will interact to form a composite or overall image. Ahmed (1991) noted that an important issue in destination image is to delineate the relationship between overall image and other components and the overall notion may be favorable or unfavorable. Keown, Jacobs and Worthley (1984) studied American tourists' perceptions of retail stores in twelve selected countries by examining the relationship among six perceptual/cognitive attributes and overall image. The authors concluded that overall impression is dependent upon individual attributes. Then, the beliefs and feelings dimensions together influence overall attitude or image. The causal linkages indicate that beliefs influence overall or composite attitude directly as well as indirectly through affect.

The conceptual and empirical perspectives from the literature led to following three hypotheses:

Hypothesis 1: Perceptual/cognitive evaluations significantly influence affective evaluations of a tourism destination.

Hypothesis 2: Perceptual/cognitive evaluations significantly influence overall image of a tourism destination.

Hypothesis 3: Affective evaluations significantly influence overall image of a tourism destination.

Information Sources. According to Woodside and Lysonski's (1989) destination choice model, marketing variables or information sources are a force which influences the formation of perceptions or cognitive evaluations but not the affective component of image. Similarly, in Um and Crompton's (1990) and Um's (1993) model of pleasure destination choice, the perceptual/cognitive evaluation of attributes (beliefs) are formed by external factors which include various information sources such as symbolic stimuli (promotional efforts of a destination through media) and social stimuli (friends and relatives' recommendations or word-of-mouth). The role of information sources in this formation is also emphasized in Fakeye and Crompton's (1991) model. Adapting from Gunn's (1972) notion of organic and induced images, their model described tourists developing organic images of a set of alternative destinations from various non-tourism information sources. With the desire to travel, they may get involved in an active information search and resort to specific information sources. As a result, they develop induced images of alternative choices which may be the same as, or substantially different from, their organic images. Gartner (1993) noted that the type and amount of external stimuli (information sources) received influence the formation of the cognitive component of image but not the affective counterpart. In this sense, the development of the perceptual/cognitive component is presented as a function of the variety (amount) and type of information sources to which tourists are exposed. Burgess (1978) hypothesized that the type, quality, and quantity of information would determine the type of image is likely to develop. The empirical results of Holbrook's (1978) study indicated that information sources do influence the cognitive (but not affective) component of image. In other words, the cognitive component plays an intervening role between information sources and the affective component (Holbrook 1978). Two hypotheses were derived from this proposition:

Hypothesis 4: The variety (amount) of information sources used significantly influence perceptual/cognitive evaluations.

Hypotheses 5: The type of information sources used significantly influences perceptual/cognitive evaluations.

Sociopsychological Motivation. People engage in tourism for different reasons or motives. Motivation is accepted as the central con-

cept in understanding tourism behavior and the destination choice process (Uysal and Hagan 1993; Weaver, McCleary, Lepisto and Damonte 1994), because they are the impelling and compelling force behind all actions (Crompton 1979b; Iso-Ahola 1982). Motivations are usually defined as sociopsychological forces that predispose an individual to opt for and participate in a touristic activity (Beard and Raghep 1983; Crandall 1980; Iso-Ahola 1982). They are also included in destination choice and image formation models as major influence factors (Stabler 1990; Um 1993; Um and Crompton 1990).

The image of a destination is highly related to touristic benefits sought (motivations). They determine the image of a destination before and after the visitation (Mill and Morrison 1992:32–56). In the destination choice process, images are formed in relation to the motivations in a conscious or unconscious way (Moutinho 1987:18). Mayo and Jarvis (1981:30) indicated that tourists' psychological motivations influence their images of destinations. Pearce (1995:173) suggested that the relationship between motivations and destination image should be explored to better understand their behavior and to enhance motivation theory. Several authors suggested that motivations are related to the affective component of image and an individual's affective image toward a destination is, to a great extent, influenced by his/her motivations (benefits sought) from the touristic experience (Dann 1996; Gartner 1993; Walmsley and Jenkins 1993). Hence, the following hypothesis was proposed:

Hypothesis 6: Tourists' sociopsychological motivations significantly influence their affective evaluations of destinations.

Demographic Variables. Most image formation and destination selection models have incorporated sociodemographic variables as conventional consumer characteristics influencing perceptions of objects, products, and destinations (Friedmann and Lessig 1986; Stabler 1990; Um and Crompton 1990; Woodside and Lysonski 1989). The consumer behavior models of Fisk (1961–62) and Sheth (1983) also recognized the sociodemographic characteristics of consumers as determinants of consumer image by including them as antecedents to cognitive processes. Although such variables as age, education, income, gender, occupation, and marital status have all been suggested as influencing perceptions and images, age and education appear to be major determinants of image. Nickel and Wertheimer (1979) studied the effects of age, education, occupation, income, marital status, and size of the family on consumer images of drugstores and found that age was the only variable affecting the process.

Walmsley and Jenkins (1993) studied affective images of several resorts in the North Coast of New South Wales, Australia. A principal component analysis indicated that affective images of a few resorts showed variations due to gender and age. Baloglu (1997)

examined image variations of the United States based on sociodemographic characteristics of West German tourists. The author found a few image differences due to age, marital status, and occupation. However, age was the most significant sociodemographic variable. Husbands (1989) investigated the relationship between perception of tourism and sociodemographic variables and found that perception among Livingstone, Zambia locals differed significantly based on only age and education variables. Stern and Krakover (1993) chose education level as one of the most important consumer characteristics and investigated the effects of education level of individuals on the relationship between cognitive, affective, and overall image. Path analysis indicated that the strength and direction (sign) of causal effects among the three constructs showed variations due to different education groups. Therefore, based on these assessments, because age and education appear to be the most influential sociodemographic variables on image, they were selected for this study. Accordingly, two more hypotheses were formed:

Hypothesis 7: Tourists' ages significantly influence their perceptual/cognitive and affective evaluations of destinations.

Hypothesis 8: Tourists' education levels significantly influence their perceptual/cognitive and affective evaluations of destinations.

Research Design

The study was conducted using an explanatory research design which advances and tests hypotheses to confirm causal relationships. The hypotheses were tested based on the data collected by a self-administered questionnaire which was mailed to respondents. They were asked to answer all of the questions that related to the variables to be measured for four different countries: Turkey, Greece, Italy, and Egypt. Each respondent was asked to answer the same set of questions for all four countries. These countries were used to test the model for validity and also to allow for image comparison among them (the latter is beyond the scope of this article).

The survey instrument consisted of four sections: questions relating to sociopsychological motivations and screening questions relating to previous experience with selected destinations; questions relating to variety and type of information sources used regarding selected destinations; questions relating to the image construct and measurement of the respondents' overall image and affective and perceptual/cognitive evaluations of destinations; and questions designed to gather demographic information. The instrument was pretested on 60 undergraduate students as well as the faculty and graduate students in a tourism program at a major university. It was also sent to three travel agents who were involved with booking

destinations selected for this study. The instrument was revised based on the results of the pretesting procedures.

People who engage in information search about destinations can be considered potential tourists and represent the population of interest for this study. Therefore, the sample for this study consisted of adults (18 years old or over) derived from a list maintained by the Turkish National Tourism Office in New York. The list consisted of 4,600 potential tourists who requested information about Turkey between the periods of November 1994 and October 1995, but who had not been to Turkey at the time they requested information. As an incentive for participating in the study, the respondents were informed that they would be eligible to win several prizes, including a round-trip airline ticket to Turkey. All sample members were asked to answer questions regarding all four countries included in the study, even though they had requested only information about Turkey. Therefore, It is possible that there could be a potential bias toward Turkey from this sampling method and response incentive. This sampling method was followed because Turkey was of particular interest to the researchers.

An important and complex issue in sampling is to determine the appropriate sample size to be used. This determination largely depends on the statistical estimating precision needed by the researcher and the number of variables. Although larger sample sizes are preferred, a number of respondents of between 200 and 400 is usually accepted as the critical sample size for multiple regression and path analysis (Hair, Anderson, Tatham and Black 1992). However, consideration was also given to the fact that this study sought to test an image development model before actual visitation to destinations and some respondents who might have visited selected destinations since requesting information would be excluded from the analysis. In fact, a screening question asking whether the respondent has visited a destination was included on the questionnaire so that those with previous experience could be excluded from the analysis. Therefore, the targeted usable sample size was set at 400. Previous destination image studies using multiple destinations and attributes had reported a response rate between 26% and 40%. Assuming a conservative response rate of about 26%, 1,530 potential tourists were surveyed to achieve the targeted sample size. A systematic random procedure was employed to draw a representative sample of 1,530 from the population of 4,600 by selecting every 3rd person ($4,600/1,530=3$).

The study was conducted during the spring of 1996. To ensure the targeted response rate, the data collection procedure was initially planned to use a modified total design method proposed by Dillman (1978) where two mailing waves were considered. The questionnaire (along with a cover letter, prize entry form, and postage-paid return envelope) was mailed on May 14, 1996. Approximately three weeks after the initial mailing, a total of 484 questionnaires were returned, representing a 31.6% response rate. As this response rate from the first wave was above the expected

minimum, the follow-up mailing (second wave) was not conducted. Elimination of unusable questionnaires resulted in a total of 448 completed responses.

The hypotheses and the model were formulated in a general nature rather than specific to four tourism destinations. Therefore, these were tested on the combined data set (pooled data) regardless of destinations. Data coding was performed according to a pooled cross-sectional design where dummy variables were used for destinations (Dielman 1988; Kliman 1981). One advantage of a pooled cross-sectional design is that it increases sample size by the number of testing units (destinations). The responses of the pooled data increased to 1,792 (448×4) because each of the 448 respondents filled out the questionnaire for all four countries.

Table 1. Variables and Sources for Scale Items Used for Measurement

Variable	Source of Scale for Measurement
Overall Image	Stern and Krakover (1993) Image Formation of a City
Perceptual/Cognitive Evaluations	Gearing, Swart and Var (1974) Touristic Attractiveness Criteria Tourism Canada (1989) Destination Brochures and Guidebooks
Affective Evaluations	Russel (1980) Russel and Pratt (1980) Russel et al (1981) Russel and Snodgrass (1987)
Variety (amount) of Information Sources	Stern and Krakover (1993) Tourism Canada (1989)
Types of Information Sources	Um and Crompton (1990) Gitelson and Crompton (1983) Hsieh and O'Leary (1993) Gunn (1972) Gartner (1993)
Age/Education	Open-ended/Number of Years Schooling
Sociopsychological Travel Motivations	Crompton (1979b); Dann (1981); Iso-Ahola (1982) Beard and Ragheb (1983) Leisure Motivation Scale Lounsbury and Franz (1990) Vacation Motivations Tourism Canada (1985, 1989) Fisher and Price (1991) International Travel Motivations Loker and Perdue (1992) Shoemaker (1989) McIntosh and Goeldner (1990) Uysal and Hagan (1993)

Since the purpose of this study was to test a model of image determinants (image formation) in the *absence of actual visitation*, previous actual visitation to selected destinations was used to screen the respondents. A subsample ($n=356$) was identified from the entire set of 448 respondents by eliminating anyone who had visited all four countries. Then, if a respondent in the remaining subsample had visited a particular country, the responses for that country were eliminated. For example, if a respondent had visited Greece but none of the other three countries, the responses to the questions about Greece were not used but the responses for the other countries were kept in the pooled data and the person represented three responses. This process resulted in a subsample of 716 responses which included only evaluations of destination countries not visited by them. These 716 responses are the base for the data analysis. In order to check non-response bias, a random sample of 39 individuals who did not reply was telephoned to compare the characteristics and opinions of respondents and non-respondents. Data were collected on sociodemographics, previous experience, affective evaluations, and overall image. No significant differences were found between the two groups based on selected variables.

Measurement of Variables

The measurement of variables is presented by working backward through the model. Table 1 lists each of the variables and the studies from where the scales used to measure each variable were drawn. To study endogenous variables, respondents were asked to rate their overall image of each country as a vacation destination on a 7-point anchor scale ranging from "very positive" to "very negative". As depicted in the model, two constructs serve as intervening variables between exogenous variables and the ultimate dependent variable (overall image): perceptual/cognitive and affective evaluations of destinations. Affective evaluation also plays an intervening role between the perceptual/cognitive evaluation and overall image.

Fifteen perceptual/cognitive evaluation items were generated from the literature review and a content analysis of the four destinations' guidebooks and brochures so that attributes selected could be applied to all the destinations included in this study. The items were then pre-tested on a sample of 60 students. One item, "ease of communication" was discarded after the pretest because it did not load on any factor grouping for all destinations. The remaining 14 items were included in the questionnaire. Respondents were asked to rate each country as a summer vacation destination on each of 14 attributes on a 5-point scale, offers very little, offers somewhat little, neither little or much, offers somewhat much, and offers very much (1 to 5, respectively).

Affective evaluations of destinations were measured using four bipolar scales (Arousing–Sleepy, Pleasant–Unpleasant, Exciting–Gloomy, and Relaxing–Distressing). A composite score of four bipolar scales provides an overall affective evaluation of a given destina-

tion (Russel and Snodgrass 1987). As suggested by Russel and his colleagues, although only two of the scales (Arousing–Sleepy and Pleasant–Unpleasant) are theoretically adequate to measure the affective image, the reliability of environmental perception can be increased by using all four scales. The reliability and validity of the scales have been proven over different languages, samples, cultures, and environment types (Baloglu and Brinberg 1997; Hanyu 1993; Russel and Snodgrass 1987; Russel et al 1981, 1989; Walmsley and Jenkins 1993; Ward and Russel 1981).

Exogenous variables were also examined. Following Stern and Krakover (1993), the variety (amount) of information sources used was measured by a variable indicating the number of institutions, services, and materials through which the respondents have seen or heard about destinations. Nine different information sources were compiled from the literature: travel agents, brochures/travel guides, friends/family members, airlines, tour operator/company, advertisements, books/movies, articles/news, and direct mail from destination. Out of this list, the variety of information score was calculated as the sum of the number of information sources used and ranged from 0 (minimum) to 9 (maximum).

Four information source categories were used to measure the importance of type of in forming impressions about selected destinations. These categories represent all individual information source items listed when measuring the variety (amount) used. The categories were derived from the literature as noted on Table 1. The four information source categories included professional advice (tour operators, travel agents, and airlines); word-of-mouth (friends, relatives, and social clubs); advertisement (print or broadcast media); and books/movies/news. Respondents were asked to rate each information category as to how important it was in forming their impressions about each destination country on a 4-point Likert type scale, ranging from “not at all important” to “very important”.

Respondents’ age was measured by seeking an open-ended answer. Education level was measured by the number of schooling years completed. As to sociopsychological motivators, the literature review of this topic revealed that no single established scale exists to measure tourists’ international motivations. Therefore, a variety of studies listed on Table 1 were drawn upon to identify 23 sociopsychological motivation items. Four items were added to assess precision and for cross-validation, which resulted in a total of 27 items. Precision of the scales is the level of “agreement among raters” (Menezes and Elbert 1979:82). These items were then tested on a sample of 45 students and faculty.

The 27 motivation items were factor analyzed using principal component analysis with a varimax rotation procedure. All factors which had an eigenvalue greater than one and factor loadings .40 or greater were retained. In the initial solution, three items did not meet the .40 criteria and were eliminated. In addition, several factors included redundant items (the attributes which were added to check the consistency and precision). To prevent unnecessary dupli-

cation, redundant items were eliminated based on the lowest item-to-total correlation. This resulted in 17 motivation items which were factor analyzed again using the same procedure and criteria. All items met the .40 cut-off point and produced six factors. The percentage of variance explained by this final solution was 70.2%. The consistency or precision of the scale was assessed by correlation coefficients, which was greater than .75 for all items added for cross-validation. Therefore, the scale was judged as having consistency and precision. The 17 sociopsychological motivations were included in the questionnaire where respondents were asked to evaluate each reason on a scale of ranging from “not at all important” to “very important”.

Study Findings

The demographic profile of the respondents indicated that gender of the respondents was almost evenly distributed, with 47.5% male and 52.5% female. The majority of the respondents were within

Table 2. Factor Analysis of Perceptual/Cognitive Items

Factor	Factor Loadings	Eigenvalue	Variance Explained
Factor I:		5.87	41.9%
Quality of Experience (.89)^a			
Standard Hygiene and Cleanliness	.84		
Quality of Infrastructure	.81		
Personal Safety	.76		
Good Nightlife and Entertainment	.70		
Suitable Accommodations	.69		
Appealing Local Food (Cuisine)	.66		
Great Beaches/Water Sports	.65		
Interesting and Friendly People	.52		
Factor II:		1.67	11.9%
Attractions (.72)			
Interesting Cultural Attractions	.86		
Interesting Historical Attractions	.85		
Beautiful Scenery/Natural Attractions	.56		
Factor III:		1.07	7.6%
Value/Environment (.56)			
Good Value for Money	.82		
Unpolluted/Unspoiled Environment	.54		
Good Climate	.50		
Total Variance Explained			61.5%

^aReliability score (Cronbach alpha) for each factor grouping is shown in parentheses.

older age brackets, with 35.1% between 50 and 64, followed by 65 years old or older (30.6%). Only 11.7% belonged to the 18–34 age group. Most of the respondents were married (62.7%). All were highly educated, 44.4% attended college, and 47.5% attended graduate school. Almost 70% of the respondents reported a yearly income

Table 3. Factor Analysis of Travel Motivation Items

Factor	Factor Loading	Eigenvalue	Variance Explained
Factor I:		4.12	24.3%
Relaxation/Escape (.80)^a			
Relieving stress and tension	.85		
Getting away from demands of everyday life	.81		
Relaxing physically and mentally	.75		
Getting away from crowds	.62		
Escaping from the routine	.62		
Factor II:		2.23	13.1%
Excitement/Adventure (.73)			
Doing exciting things	.82		
Finding thrills and excitement	.82		
Being adventurous	.63		
Having fun, being entertained	.57		
Factor III:		1.67	9.8%
Knowledge (.64)			
Learning new things, increasing my knowledge	.79		
Experiencing different cultures and ways of life	.73		
Enriching myself intellectually	.66		
Experiencing new/different places	.58		
Factor IV:		1.50	8.8%
Social (.64)			
Meeting people with similar interests	.83		
Developing close friendships	.76		
Factor V:		1.09	6.4%
Prestige (.61)			
Going places my friends have not been	.87		
Telling my friends about the trip	.71		
Total Variance Explained			62.4%

^aReliability score (Cronbach alpha) for each factor grouping is shown in parentheses.

of \$50,000 or more. Twenty-eight respondents earned an income of \$100,000 or more, followed by another 24.0% who earned between \$75,000 and \$99,999. Only 8.2% reported a yearly income under \$25,000.

Factor Analysis. An exploratory factor analysis was performed on sociopsychological tourism motivations and perceptual/cognitive image items. The factor analysis on perceptual/cognitive items was conducted on the pooled data set from the responses to all four destinations. Principal component and varimax rotation procedures were used to identify orthogonal factor dimensions. The latent root criterion of 1.0 was utilized for factor extraction and factor loadings of .40 were utilized for item inclusion (Hair et al 1992; Nunnally 1978). All motivation (17 items) and perceptual/cognitive (14 items) items from the questionnaire met this cut-off point. Factor scores were computed by taking the average of items within each factor.

Table 4. Descriptive Statistics for Latent and Observed Variables in the Model^a

Variables	Construct			
	Mean	SD	Reliability ^b	No. of Items
1. Variety of Information Sources (VIS)	3.75	2.01		9
2. Professional Advice (TIS1)	3.00	.94		
3. Word-of-Mouth (TIS2)	3.29	.86		
4. Advertisement (TIS3)	2.65	.85		
5. Books/Movies/News (TIS4)	3.04	.86		
6. Age (AGE)	52.55	14.68		
7. Education (EDUC)	16.70	2.83		
8. Relaxation/Escape (MOT1)	2.87	.70	.80	5
9. Excitement/Adventure (MOT2)	2.92	.64	.73	4
10. Knowledge (MOT3)	3.81	.30	.64	4
11. Social (MOT4)	2.34	.75	.64	2
12. Prestige (MOT5)	2.02	.76	.61	2
13. Quality of Experience (COG1)	3.34	.80	.89	8
14. Attractions (COG2)	4.59	.57	.72	3
15. Value/Environment (COG3)	3.55	.80	.56	3
16. Affect (AFF)	5.24	1.44	.88	4
17. Overall Image (OI)	5.05	1.87		

^a Scale: *Exogenous Variables*. Variety (amount) of Information Sources (VIS) range from 1 to 9 (summative score); Type of Information Sources (TIS1–TIS4): 1=Not At All Important, 4=Very Important; Age (AGE): Open-Ended; Education (EDUC) range from 1 to 21; Socio-Psychological Motivations (MOT1–MOT5): 1=Not At All Important, 4=Very Important. *Endogenous Variables*. Perceptual/Cognitive Evaluations (COG1–COG3): 1=Offers Very Little, 5=Offers Very Much; Affect (AFF) ranges from 1 to 7 (the higher the score, the more positive the affect); Overall Image (OI): 1=Very Negative, 7=Very Positive.

^b Construct reliability refers to the dimensions identified by factor analysis of socio-psychological motivations and perceptual/cognitive evaluations.

The factor analysis of the 14 perceptual/cognitive items from the questionnaire produced three factors and explained 61.5% of the variance (Table 2). The factors were labeled as quality of experience, attractions, and value/environment. The 17-motivation items from the factor analysis resulted in five groupings and explained 62.4% of the variance (Table 3). All of the factor loadings were greater than .60, indicating a good correlation between the items and the grouping they belong to. Factors were labeled based on highly loaded items and the common characteristics of items they included. Their labels are relaxation/escape, excitement/adventure, knowledge, social, and prestige.

Table 4 provides a summary of all the variables used to test the model along with their abbreviations. A total of 17 variables comprise the model. Some of the variables were derived from the factor analyses of the motivational and perceptual/cognitive items and have multiple items measuring each variable as is noted in this table. Reliability measures are also listed where appropriate. Internal consistency between the items in the measures was estimated using Cronbach's coefficient alpha. This is the most widely used reliability measure to estimate the degree to which the items on a measure are representative of the domain of the construct being measured. The table shows internal reliability estimates for perceptual/cognitive evaluations, sociopsychological motivations, and affect. A coefficient alpha of at least .70 is usually considered reliable. Four scales could not meet this criteria as shown in the table: knowledge (MOT3), social (MOT4), prestige (MOT5) motivations and value/environment (COG3). However, their reliability scores from the pretest were also the same. It should be noted that these lower alpha values are associated with those scales that included a smaller number of items. As the number of items increases, the alpha value also increases. Therefore, the acceptable alpha limit is as low as .60 or .50 for scales consisting of a small number of items (Carmines and Zeller 1979; Pedhazur and Schmelkin 1991). Alpha values ranged from .72 to .89 for other scales.

Path Analysis. The dynamic nature of image was examined in a holistic manner by integrating variables in a path-analytic model. To do this, each dependent (endogenous) variable was regressed on all independent (exogenous) variables that directly impinge on it. Variety (amount) of information sources used (VIS), the type of information source used (TIS1–TIS4), age (AGE), education (EDUC) and socio-psychological travel motivations (MOT1–MOT5) are the primary causal (exogenous) variables. Perceptual/cognitive evaluations (COG1–COG3) and affective evaluation (AFF) are both exogenous and endogenous variables that mediate the relationship between overall image (OI) (ultimate endogenous variable) and exogenous variables in the model. At the same time, perceptual cognitive evaluations mediate the relationships between some exogenous variables and affective evaluations.

The overall pattern of the model was scrutinized by examining causal relationships among variables (direct, indirect, and total effects of variables and the mediating roles of perceptual/cognitive and affective evaluations in forming overall image of destinations). The direct effects of exogenous variables on overall image were also explored to summarize the overall pattern of the proposed model and validate the role of mediating variables (perceptual/cognitive and affective evaluations) in the path model. Further, the data were examined for several assumptions of path analysis such as normality, linearity, homoskedasticity, and multicollinearity (Pedhazur 1982; Hair et al 1992). Residual plots, normal probability plots, descriptive statistics (mean standard deviation, mode and median), frequencies (skewness values), and correlations were used to detect any violation of assumptions. In addition, normality was checked by examining both residual plots of regression analysis and normal probability plots. The residual values did not show any substantial departure from the straight diagonal line, indicating a normal distribution. Skewness values did not fall outside the range of -1 to 1 . Values outside these limits indicate a skewed distribution.

Linearity was checked by examining the residual plots of linear regression and curve estimation. No nonlinear pattern was found between dependent and independent variables. Moreover, the assumption of homoskedasticity or equal variance is critical to linear regression. This assumption was checked by both residual plots. If the residuals increase or decrease with values of independent variables or with predicted variables, then the constant variance assumption is violated. No pattern of such relationships were found and the data were judged homoskedastic. Possible multicollinearity problems were also checked by examining correlations among independent variables in and diagnostic tests such as variance inflation factor and condition indices. No violation of multicollinearity was detected.

Support for Hypotheses. Table 5 presents a summary of the estimated causal coefficients of the path analysis (discussed later). The level of support for each hypothesis is based on the following criteria:

Path Coefficient of: 000–.05 indicates weak support
 .051–.30 indicates moderate support.
 .301–1.00 indicates strong support

H1: Perceptual/cognitive evaluations significantly influence affective evaluations of a destination. The path coefficients (shown in Table 5) from quality of experience (COG1=.370), attractions (COG2=.170), and value/environment (COG3=.290) to affect were significant and strong for quality of experience and moderate for attractions and value/environment. This provides

Table 5. Estimated Path Coefficients for Hypothesized Relationships (N=716)

Endogenous Variables	Exogenous Variables					
	COG1	COG2	COG3	AFF	OI	OI (Direct Effects) ^b
Variety (amount) of Information Sources (VIS)	.277 ^a	.295 ^a	.194 ^a			.044
Professional Advice (TIS1)	.049	.013	-.068			-.041
Word-of-Mouth (TIS2)	.134 ^a	.108 ^a	.170 ^a			.069 ^a
Advertisement (TIS3)	.053	-.005	.115 ^a			.073 ^a
Books/Movies/News (TIS4)	.030	.018	.005			-.034
Age (AGE)	-.118 ^a	-.029	-.142 ^a			-.057
Education (EDUC)	-.095	-.016	-.138 ^a			-.034
Relaxation/Escape (MOT1)				-.068 ^a		-.036
Excitement/Adventure (MOT2)				.057		.039
Knowledge (MOT3)				.040		-.078 ^a
Social (MOT4)				.063		-.100 ^a
Prestige (MOT5)				.000		.066 ^a
Quality of Experience (COG1)				.370 ^a	.073 ^a	.055
Attractions (COG2)				.170 ^a	-.039	-.034
Value/Environment (COG3)				.290 ^a	.133 ^a	.095 ^a
Affect (AFF)					.703 ^a	.717 ^a
R ²	.140	.106	.147	.502	.675	.715

^aSignificant at .05 or better probability level.

^bThe last column shows the path coefficients (effects)

when overall image was regressed on all exogenous and endogenous variables in the model (refer to Figure 2 for paths).

strong support for H1 indicating that perceptual/cognitive evaluations positively influence affective evaluations of destinations.

H2: Perceptual/cognitive evaluations significantly influence overall image of a destination. Moderate support was provided for this hypothesis as quality of experience (COG1=.073) and value/environment (COG3=.133) positively influenced overall image (OI).

H3: Affective evaluations significantly influence overall image of a destination. This hypothesis was strongly supported with a path coefficient of .703 from affect (AFF) to overall image (OI).

H4: The variety (amount) of information sources used significantly influences perceptual/cognitive evaluations. The results provided moderate support for H4 with all path coefficients from variety of information being significantly related to all three perceptual/cognitive measures (COG1=.277; COG2=.295; COG3=.194).

H5: The type of information sources used significantly influences perceptual/cognitive evaluations. The path analysis revealed that only word-of-mouth had significant and moderate path coefficients with all three perceptual/cognitive variables

(COG1=.134; COG2=.108; COG3=.170). Advertisement was significant only with COG3=.115), while the other two information factors showed no significant path coefficients.

H6: Tourists' sociopsychological motivations significantly influence their affective evaluations of destinations. This hypothesis was moderately supported, but only relaxation/escape being significant (MOT1=-.068).

H7: Tourists' ages significantly influence their perceptual/cognitive and affective evaluations of destinations. The coefficient between age and affective evaluations was not significant at all so that the only support was for the relationship between age and perceptual/cognitive, with age having an inverse and moderate relationship with quality of experience (COG1=-.118) and value/environment (COG3=-.142).

H8: Tourists' education levels significantly influence their perceptual/cognitive and affective evaluations of destinations. As with H7, this hypothesis had only partial support. There was no support for that suggestion that education influences affect and a moderate relationship between education and perceptual/cognitive evaluations was found with only value/environment (COG3=-.138) having a significant path coefficient.

Assessment of the Path Model. Based on explained causalities (variabilities), the explanatory power of the overall model, where final outcome was overall image, was strong. Primary causal variables had moderate explanatory power where perceptual/cognitive and affective evaluations were dependent variables. The R^2 values on Table 5 show variety (amount) of information sources, type of information sources and age explained 14.0% of the variability in quality of experience (COG1); variety and type of information sources explained 10.6% of the variability in attractions (COG2); and variety and type of information sources, age, and education explained 14.7% of the variability in value/environment (COG3). Sociopsychological motivations and perceptual/cognitive evaluations explained 50.2% of the variability in affect (AFF) where a large portion of that variability was attributed to perceptual/cognitive evaluations. Perceptual/cognitive evaluations and affective evaluations explained 67.5% of the variability in overall image, the ultimate dependent variable in the model. A large portion of this variability resulted from affective evaluations.

To summarize the overall pattern of the proposed model and validate the role of mediating variables (perceptual/cognitive and affective evaluations) in the path model, direct effects of exogenous variables on overall image were explored by regressing overall image on all variables preceding it in the model (Table 6, last column). This allowed a comparison of simple (direct) and compound (indirect) effects of the exogenous variables on the overall image. As is shown in this table, type of information sources, motivations,

Table 6. Decomposition of Causal Effects into Direct, Indirect, and Total Effects (N=716)^a

Exogenous Variables	Endogenous Variables					
	AFF			OI		
	DE	IE	TE	DE	IE	TE
Variety (amount) of Information Sources (VIS)		.21	.21		.17	.17
Word-of-Mouth (TIS2)		.12	.12	.07	.10	.17
Advertisement (TIS3)		.05	.05	.07	.05	.12
Age (AGE)		-.09	-.09		-.06	-.06
Education (EDUC)		-.04	-.04		-.04	-.04
Relaxation/Escape (MOT1)	-.07		-.07		-.05	-.05
Knowledge (MOT3)				-.08		-.08
Social (MOT4)				-.10		-.10
Prestige (MOT5)				.07		.07
Quality of Experience (COG1)	.37		.37		.26	.26
Attractions (COG2)	.17		.17		.12	.12
Value/Environment (COG3)	.29		.29	.10	.21	.31
Affect				.72		.72

^a Note: Table shows only those effects significant at .05 or better probability level. DE, Direct effect; IE, Indirect effect; TE, Total effect (DE+IE).

perceptual/cognitive, and affective evaluations explained 71.5% of the variability in overall image, which is a very small increase from the 67.5% explained by perceptual/cognitive and affective evaluations only.

Significant effects on overall image were found for word-of-mouth (TIS2=.069) and advertisement (TIS3=.073). These effects were moderate and positive. Three tourism motivation items, knowledge (MOT3=-.078), social (MOT4=-.100), and prestige (MOT5=.066) moderately and significantly influenced the overall image. The effects of knowledge and social motivations on overall image was negative and the effect of prestige was positive. The only significant cognitive variable, environment (COG3=.095), moderately influenced overall image. The path coefficient from affect (AFF=.717) to overall image was strong and positive.

Model Consistency. The consistency of the model was checked by dividing the sample into halves and testing the model separately. All relationships except the effects of age on perceptual/cognitive evaluations were consistent between the two groups, which were also consistent with the path model tested on the whole sample data. However, although the effects of perceptual/cognitive evaluations on overall image (H2) were consistent between the two halves, they were not consistent with the model tested on the whole sample. In other words, these effects were not supported by either half but were weakly supported on the whole sample.

To assess the predictive validity of image measures, visitation intentions was regressed on perceptual/cognitive evaluations (COG1–COG3), affect (AFF), and overall image (OI) separately. Quality of experience (COG1), attractions (COG2), and value/environment (COG3) positively influenced visitation intentions at the $P < .01$ level, explaining 28% of the variability in visitation intention. Affect and overall image also positively influenced visitation intentions at the $P < .01$ level where affect explained 21% of the variability and overall image explained 17% of the variability in visitation intentions.

Decomposition of Path Coefficients. Direct effects in recursive path models are estimated by partial regression coefficients, referred to as direct path coefficients. The indirect effect of each variable is the sum of the products of direct path coefficients from an exogenous variable through intervening variables to an endogenous variable. According to the hypothesized path model, only overall image would be subject to both direct and indirect effects from other variables in the model. Affect, on the other hand, would only be influenced indirectly by variety and type of information sources through perceptual/cognitive evaluations. Because age and/or education did not significantly influence affect, they could not have both direct and indirect effects on it. However, they have indirect effects on affect through perceptual/cognitive evaluations. All other relationships can be explained by direct path coefficients.

The indirect effects are calculated by multiplying the path coefficients from an exogenous to an endogenous variable through intervening ones. For example, variety (amount) of information sources (VIS) indirectly influences affect (AFF) through perceptual/cognitive evaluations (COG1–COG3) (Figure 2). The indirect effect of variety (amount) of information sources (VIS) on affect (AFF) can be calculated as the path coefficient between VIS and COG1 times the path coefficient between COG1 and AFF, *plus* the path coefficient between VIS and COG2 times the path coefficient between COG2 and AFF, *plus* the path coefficient between VIS and COG3 times the path coefficient between COG3 and AFF $[(.277)(.370) + (.295)(.170) + (.194)(.290) = .21]$.

Table 6 reports the resulting direct, indirect, and total effect (sum of direct and indirect effects) of all relevant variables on overall image and affect. For example, the variables in Figure 2 called variety (amount) of information sources (.21), word-of-mouth (.12), advertisement (.05), age (−.09), and education (−.04) indirectly influence affect through quality of experience, attractions, and/or value/environment, whereas direct effects come from relaxation/escape (−.07), quality of experience (.37), and attractions (.17). This table also shows that most variables had indirect effects on overall image through perceptual/cognitive and affective evaluations rather than direct effects. Quality of experience (.26), attractions (.12), and value/environment (.21) had positive indirect effects through affect on overall image. Although value/environment had a

direct effect on affect (.10), its indirect effect through affect (.21) was greater than its direct effect. This result supported the mediating role of affect between perceptual/cognitive evaluations and overall image.

CONCLUSION

The findings of the path analysis revealed the overall pattern of the model and indicated that variety (amount) of information sources, type of information sources, age, and education influence perceptual/cognitive evaluations. These and sociopsychological tourism motivations together influence affect. However, the effects of perceptual/cognitive evaluations on affect were much stronger than the effects of travel motivations.

It is important to note that the hypothesized model proposed that variety and type of information sources influence only perceptual/cognitive evaluations, while sociopsychological motivations influence only affect. An examination of the correlation matrix indicated that variety and type of information sources had a significant relationship with affect and sociopsychological motivations had a significant relationship with perceptual/cognitive evaluations. These relationships were not hypothesized in the original model and thus not examined because of lack of support from the literature. If they had been investigated, it would have been possible to compare direct and indirect effects of variety and type of information sources and sociopsychological motivations on affect. This would further validate the intervening role of perceptual/cognitive evaluations between exogenous variables and affect. Significantly, overall image is more likely influenced by affect than perceptual/cognitive evaluations and affect together. In other words, affect is more likely to serve as an intervening variable between perceptual/cognitive evaluations and overall image.

It should be noted here that the sample was homogenous in terms of characteristics. This may confound the effects of personal variables in the structural model. Therefore, the findings related to age, education, and sociopsychological motivations in the model should be viewed under this limitation. Clearly, future research is needed to further clarify the role of tourist characteristics in image formation from a diverse sample. The findings are limited to unidirectional influences among the variables in the model (a recursive causal modeling) because reciprocal (two-way) relationships among the variables were not studied (a non-recursive causal modeling). Findings are also limited to the set of destination countries included in the study. Additional research can validate the findings of this study, with the relationships among the model variables may be extended to other international destinations, or local destinations, to further assess the external validity of the findings.

Another limitation is related to seasonal image of the destinations. In this study, the respondents were asked to evaluate the destinations as summer vacation destinations. This would be a poss-

ible explanation of negative effects of age and education in the model because the sample consisted of mostly senior tourists. It is also possible that some sample members are overrepresented in the pooled data set. A respondent who has not been to one destination country was counted once while a respondent who has not been to any of the four was counted four times. An ideal case would be a selection of subsample who have been to none of the four destinations.

This study has theoretical and practical implications. From a theoretical standpoint, the study developed and tested a conceptual model of the determinants of destination image. It added to the existing knowledge by providing empirical evidence for the elements contributing to the development of tourism destination images. It was found that a destination image was formed by both consumer characteristics and stimulus factors. Its uniqueness was to illustrate the dynamic structure of image and simultaneous treatment of the elements contributing to destination image development. In this sense, the study shed some light on image formation theory. It empirically demonstrated that the elements that influence destination images are multi-dimensional.

Another important implication of this study is that the formation of destination images is dependent on the different roles played by the factors in the process. Variety (amount) and type of information sources used about destinations and tourists' sociodemographic characteristics influence the perceptions and cognitions of destination attributes. Somewhere in the process these perceptions, together with travelers' socio-psychological motivations, form feelings towards destinations. These then mostly form the overall image of tourism destinations. An interesting point to note is that the perceptual/cognitive evaluations are formed as an amalgam of both positive determinants (variety and type of information sources used) and inverse determinants (age and education). In other word, they are impacted negatively by increasing levels of age and education.

Some relationships tested confirmed the findings of Holbrook (1978) regarding the relationship between information sources and perceptual/cognitive evaluations and those of Stern and Krakover (1993) in terms of the relationship among perceptual/cognitive, affective, and overall image. The role of consumer characteristics in image formation should further be studied to understand whether they are *determining*, as they were treated in this study, or *conditioning* variables (i.e., conditioning the relationship between stimuli variables and image).

Destination marketers need a better understanding of how an image is formed and what determines the process. From a practical standpoint, this study provides important implications for strategic image management and development efforts. Because of the positive contribution of the variety (amount) of information sources to perceptual/cognitive evaluations, destinations should find ways to make tourists use multiple information sources. Countries can

assess their promotional channels to develop more effective promotional programs. This study demonstrated that different types of information sources have varying degrees of effect on perceptual/cognitive evaluations. Word-of-mouth recommendations from friends and relatives was the most important source in forming touristic images. Therefore, destinations should keep in mind that providing a pleasant experience for their tourists has a major effect on the development of positive images for non-visitors. The sociodemographic variables, age, and education need also to be considered, but have a lesser influence than those relating to information.

The motivation factors in the model need to be carefully studied so that marketers can incorporate significant motivators into communications. The results showed that knowledge, prestige, and social motivations directly impact image and excitement/adventure has an indirect impact. Advertising themes can be developed to effectively create and maintain destination image by appealing to these motivations.

Countries seeking to increase their tourism share should consider the characteristics of their target markets and tailor their image development and positioning efforts to specific sociodemographic and motivation segments.

As the model test revealed, intervening variables, especially affect and to a lesser extent perceptual/cognitive ones, have a strong influence on image. Marketers should understand that to influence affective evaluations of their destinations, both destination attributes and tourists' motivations should be taken into account. Destinations spend considerable time and money to create and enhance a favorable image. Focusing on the most important variables as revealed by the relationships in the model will provide more efficiency in tourism demand stimulation expenditures and more effectiveness in attracting tourists who are evaluating new potential destinations. ■

Acknowledgments—The authors gratefully acknowledge Mersin School of Tourism and Hotel Management, Turkey; Turkish Airlines and Turkish Tourism Office, New York; and three US-based operators for providing funding and support for the study.

REFERENCES

- Ahmed, Z. U.
1991 The Influence of the Components of a State's Tourist Image on Product Positioning Strategy. *Tourism Management* 12:331–340.
- Anand, P., M. B. Holbrook, and D. Stephens
1988 The Formation of Affective Judgments: The Cognitive-Affective Model versus the Independence Hypothesis. *Journal of Consumer Research* 15:386–391.
- Assael, H.
1984 *Consumer Behavior and Marketing Action*. Boston: Kent.
- Baloglu, S.
1997 The Relationship between Destination Images and Sociodemographic and Trip Characteristics of International Travelers. *Journal of Vacation Marketing* 3:221–233.

- 1998 An Empirical Investigation of Attitude Theory for Tourism Destinations: A Comparison of Visitors and Non-visitors. *Journal of Hospitality and Tourism Research* 22:221–224.
- Baloglu, S., and D. Brinberg
1997 Affective Images of Tourism Destinations. *Journal of Travel Research* 35(4):11–15.
- Beard, J. G., and M. G. Raghep
1983 Measuring Leisure Motivation. *Journal of Leisure Research* 15:219–228.
- Boulding, K. E.
1956 *The Image: Knowledge and Life in Society*. Ann Arbor MI: University of Michigan Press.
- Brokaw, S.C.
1990 An Investigation of Jewelry Store Image Structure. Unpublished Ph.D. dissertation. Florida State University, Tallahassee.
- Burgess, J. A.
1978 Image and Identity. Occasional Papers in Geography, No. 23, University of Hull Publications.
- Carmines, E.G. and R.A. Zeller
1979 Reliability and Validity Assessment. Quantitative Applications in the Social Sciences Series, No. 17. Newbury Park CA: Sage.
- Chon, K.
1990 Traveler Destination Image Modification Process and its Marketing Implications. *Developments in Marketing Science* 13:480–482.
- Crandall, R.
1980 Motivations for Leisure. *Journal of Leisure Research* 2:45–53.
- Crompton, J. L.
1979a An Assessment of the Image of Mexico as a Vacation Destination and the Influence of Geographical Location Upon That Image. *Journal of Travel Research* 17(4):18–23.
1979b Motivations for Pleasure Vacation. *Annals of Tourism Research* 6:408–424.
- Crompton, J. L., and P. K. Ankomah
1993 Choice Set Propositions in Destination Decisions. *Annals of Tourism Research* 20:461–476.
- Dann, G.
1981 Tourism Motivation: An Appraisal. *Annals of Tourism Research* 8:187–219.
1996 Tourist Images of a Destination: An Alternative Analysis. In *Recent Advances in Tourism Marketing Research*, D. R. Fesenmaier, J. T. O'Leary and M. Uysal, eds., pp. 41–55. New York: The Haworth Press.
- Dichter, E.
1985 What is in an Image? *Journal of Consumer Marketing* 2:39–52.
- Dielman, T. E.
1988 *Pooled Cross-Sectional and Time Series Data Analysis*. New York: Dekker.
- Dillman, D. A.
1978 *Mail and Telephone Surveys*. New York: Wiley.
- Dobni, D., and G. M. Zinkhan
1990 In Search of Brand Image: A Foundation Analysis. *Advances in Consumer Research* 17:110–119.
- Driscoll, A., R. Lawson, and B. Niven
1994 Measuring Tourist Destination Perceptions. *Annals of Tourism Research* 21:499–511.
- Echtner, C. M., and B. Ritchie
1993 The Measurement of Destination Image: An Empirical Assessment. *Journal of Travel Research* 31(4):3–13.
- Fakeye, P. C., and J. L. Crompton
1991 Image Differences between Prospective, First-Time, and Repeat Visitors to the Lower Rio Grande Valley. *Journal of Travel Research* 30(2):10–16.
- Fisher, R., and L. Price
1991 International Pleasure Travel Motivations and Post-Vacation Cultural Attitude Change. *Journal of Leisure Research* 23:193–208.

- Fisk, G.
1961–1962 A Conceptual Model for Studying Consumer Image. *Journal of Retailing* 37(4):1–8.
- Fridgen, J. D.
1987 Use of Cognitive Maps to Determine Perceived Tourism Region. *Leisure Sciences* 9:101–117.
- Friedmann, R., and V. P. Lessig
1986 A Framework of Psychological Meaning of Products. *Advances in Consumer Research* 13:338–342.
- Gartner, W. C.
1986 Temporal Influences on Image Change. *Annals of Tourism Research* 13:635–644.
1989 Tourism Image: Attribute Measurement of State Tourism Products Using Multidimensional Techniques. *Journal of Travel Research* 28(2):16–20.
1993 Image Formation Process. *In* *Communication and Channel Systems in Tourism Marketing*, M. Uysal and D. R. Fesenmaier, eds., pp. 191–215. New York: Haworth Press.
- Gartner, W. C., and J. D. Hunt
1987 An Analysis of State Image Change over a Twelve-Year Period (1971–1983). *Journal of Travel Research* 26(2):15–19.
- Gearing, C., W. Swart, and T. Var
1974 Establishing a Measure of Touristic Attractiveness. *Journal of Travel Research* 12:1–8.
- Genereux, R. L., L. M. Ward, and J. A. Russel
1983 The Behavioral Component in the Meaning of Places. *Environmental Psychology* 3:43–55.
- Gitelson, R. J., and J. L. Crompton
1983 The Planning Horizons and Sources of Information used by Pleasure Vacationers. *Journal of Travel Research* 23(3):2–7.
- Goodall, B.
1990 How Tourists Choose Their Holidays: An Analytical Framework. *In* *Marketing in the Tourism Industry: The Promotion of Destination Regions*, B. Goodall and G. Ashworth, eds., pp. 1–17. London: Routledge.
- Goodrich, J. N.
1978 The Relationship between Preferences for and Perceptions of Vacation Destinations. *Journal of Travel Research* 17(2):8–13.
- Gunn, C.
1972 *Vacationscape: Designing Tourist Regions*. Austin: Bureau of Business Research, University of Texas.
- Hair, J. F., R. E. Anderson, R. L. Tatham, and W. C. Black
1992 *Multivariate Data Analysis with Readings*. New York: Macmillan.
- Han, C. M.
1989 Country Image: Halo or Summary Construct. *Journal of Marketing Research* 26:222–229.
- Hanyu, K.
1993 The Affective Meaning of Tokyo: Verbal and Nonverbal Approaches. *Journal of Environmental Psychology* 13:161–172.
- Holbrook, M. B.
1978 Beyond Attitude Structure: Toward the Informational Determinants of Attitude. *Journal of Marketing Research* 15(November):545–556.
- Hsieh, S., and J. T. O'Leary
1993 Communication Channels to Segment Pleasure Travelers. *In* *Communication and Channel Systems in Tourism Marketing*, M. Uysal and D. R. Fesenmaier, eds., pp. 57–75. New York: Haworth Press.
- Hu, Y., and B. Ritchie
1993 Measuring Destination Attractiveness: A Contextual Approach. *Journal of Travel Research* 32(2):25–34.
- Hunt, J. D.
1975 Image as a Factor in Tourism Development. *Journal of Travel Research* 13(3):1–7.

- Husbands, W.
1989 Social Status and Perception of Tourism in Zambia. *Annals of Tourism Research* 16:237–253.
- Iso-Ahola, S. E.
1982 Toward a Social Psychological Theory of Tourism Motivation: A Rejoinder. *Annals of Tourism Research* 9:256–262.
- Javalgi, R. G., E. G. Thomas, and S. R. Rao
1992 U.S. Pleasure Travelers' Perceptions of Selected European Destinations. *European Journal of Marketing* 26(7):45–64.
- Keown, C., L. Jacobs, and R. Worthley
1984 American Tourists' Perceptions of Retail Stores in 12 Selected Countries. *Journal of Travel Research* 22(3):26–30.
- Kliman, M. L.
1981 A Quantitative Analysis of Canadian Overseas Tourism. *Transportation Research* 15A:487–497.
- Kotler, P., D. H. Haider, and I. Rein
1993 *Marketing Places: Attracting Investment, Industry, and Tourism to Cities, States, and Nations*. New York: The Free Press.
- Lawson, F., and M. Baud-Bovy
1977 *Tourism and Recreational Development*. London: Architectural Press.
- Loker, L., and R. Perdue
1992 A Benefit-based Segmentation of a Nonresident Summer Travel Market. *Journal of Travel Research* 31(1):30–35.
- Lounsbury, J. W., and C. P. G. Franz
1990 Vacation Discrepancy: A Leisure Motivation Approach. *Psychological Reports* 66:699–702.
- Lynch, K.
1960 *The Image of the City*. Cambridge MA: Massachusetts Institute of Technology.
- MacKay, K. J., and D. R. Fesenmaier
1997 Pictorial Element of Destination in Image Formation. *Annals of Tourism Research* 24:537–565.
- Martineau, P.
1958 The Personality of the Retail Store. *Harvard Business Review* 36(January–February):47–55.
- Mayo, E. J.
1973 Regional Images and Regional Travel Destination. In *Proceedings of the Fourth Annual Conference of Travel and Tourism Research Association*, pp. 211–217. Salt Lake City UT: TTRA.
- Mayo, E. J., and L. P. Jarvis
1981 *The Psychology of Leisure Travel*. Boston: CBI.
- Mazursky, D., and J. Jacoby
1986 Exploring the Development of Store Images. *Journal of Retailing* 62(2):145–165.
- McIntosh, W. R., and C. R. Goeldner
1990 *Tourism: Principles, Practices, Philosophies* (6th ed.). New York: Wiley.
- Menezes, D., and N. F. Elbert
1979 Alternative Semantic Scaling Formats for Measuring Store Image: An Evaluation. *Journal of Marketing Research* 16(February):80–87.
- Mercer, D.
1971 The Role of Perception in the Recreation Experience: A Review and Discussion. *Journal of Leisure Research* 3:261–276.
- Mill, R. C., and A. M. Morrison
1992 *The Tourism System: An Introductory Text*. Englewood NJ: Prentice–Hall.
- Milman, A., and A. Pizam
1995 The Role of Awareness and Familiarity with a Destination: The Central Florida Case. *Journal of Travel Research* 33(3):21–27.
- Moutinho, L.
1987 Consumer Behavior in Tourism. *European Journal of Marketing* 21(10):5–44.

- Myers, J. G.
1968 Consumer Image and Attitude. Berkeley: Institute of Business and Economic Research, University of California, Berkeley.
- Nickel, P., and A. I. Wertheimer
1979 Factors Affecting Consumers' Images and Choices of Drugstores. *Journal of Retailing* 55(2):71-78.
- Nunnally, J. C.
1978 Psychometric Theory. New York: McGraw-Hill.
- Oxenfeldt, A.R.
1974-75 Developing a Favorable Price-Quality Image. *Journal of Retailing* 50(4):8-14, 115.
- Pearce, P. L.
1982 Perceived Changes in Holiday Destinations. *Annals of Tourism Research* 9:145-164.
1995 Pleasure Travel Motivation. *In* *Tourism: Principles, Practices, Philosophies* (7th ed.), R. W. McIntosh, C. R. Goeldner and J. R. Brent Ritchie, eds., pp. 167-178. New York: Wiley.
- Pedhazur, E. J.
1982 Multiple Regression in Behavioral Research (2nd ed.). Fort Worth: Harcourt Brace College Publishers.
- Pedhazur, E. J., and L. P. Schmelkin
1991 Measurement, Design, and Analysis: An Integrated Approach. Hillsdale NJ: Erlbaum.
- Phelps, A.
1986 Holiday Destination Image: The Problem of Assessment. *Tourism Management* 7:168-180.
- Proshonsky, H. M., A. K. Fabian, and R. Kaminoff
1983 Place-Identity: Physical World Socialization of the Self. *Journal of Environmental Psychology* 3:57-83.
- Reibstein, D. J., C. H. Lovelock, and R. P. Dobson
1980 The Direction of Causality Between Perceptions, Affect, and Behavior: An Application to Travel Behavior. *Journal of Consumer Research* 6:370-376.
- Russel, J. A.
1980 A Circumplex Model of Affect. *Journal of Personality and Social Psychology* 39:1161-1178.
- Russel, J. A., M. Lewicka, and T. Niit
1989 A Cross-Cultural Study of a Circumplex model of Affect. *Journal of Personality and Social Psychology* 57:848-856.
- Russel, J. A., and G. Pratt
1980 A Description of Affective Quality Attributed to Environment. *Journal of Personality and Social Psychology* 38:311-322.
- Russel, J. A., and J. Snodgrass
1987 Emotion and Environment. *In* *Handbook of Environmental Psychology*, D. Stockols and I. Altman, eds., pp. 245-280. New York: Wiley.
- Russel, J. A., L. M. Ward, and G. Pratt
1981 Affective Quality Attributed to Environments: A Factor Analytic Study. *Environment and Behavior* 13:259-288.
- Scott, D. R., C. D. Schewe, and D. G. Frederick
1978 A Multi-brand/Multi-attribute Model of Tourist State Choice. *Journal of Travel Research* 17(3):23-29.
- Sheth, J. N.
1983 An Integrative Theory of Patronage Preference and Behavior. *In* *Patronage Behavior and Retail Management*, W. R. Darden and R. F. Lusch, eds. New York: Elsevier Science.
- Shoemaker, S.
1989 Segmentation of the Senior Pleasure Travel Market. *Journal of Travel Research* 27(3):14-21.
- Stabler, M. J.
1990 The Image of Destination Regions: Theoretical and Empirical Aspects. *In* *Marketing in the Tourism Industry: The Promotion of Destination Regions*, B. Goodall and G. Ashworth, eds., pp. 133-161. London: Routledge.

- Stern, E., and S. Krakover
 1993 The Formation of a Composite Urban Image. *Geographical Analysis* 25(2):130-146.
- Tourism Canada
 1985 U.S. Pleasure Travel Market in Canada. Toronto: Market Facts of Canada.
 1989 Pleasure Travel Markets to North America. Montreal: Market Facts of Canada.
- Tuan, Y.
 1975 Images and Mental Maps. *Annals of the Association of American Geographers* 65:205-213.
- Um, S.
 1993 Pleasure Travel Destination Choice. In *VNR's Encyclopedia of Hospitality and Tourism*, M. Khan, M. Olsen and T. Var, eds., pp. 811-821. New York: Van Nostrand Reinhold.
- Um, S., and J. L. Crompton
 1990 Attitude Determinants in Tourism Destination Choice. *Annals of Tourism Research* 17:432-448.
- Uysal, M., and L. A. R. Hagan
 1993 Motivation of Pleasure Travel and Tourism. In *VNR's Encyclopedia of Hospitality and Tourism*, M. Khan, M. Olsen and T. Var, eds., pp. 798-810. New York: Van Nostrand Reinhold.
- Walmsley, D. J., and J. M. Jenkins
 1993 Appraisive Images of Tourist Areas: Application of Personal Construct. *Australian Geographer* 24(2):1-13.
- Ward, L. M., and J. A. Russel
 1981 The Psychological Representation of Molar Physical Environments. *Journal of Experimental Psychology: General* 110(2):121-152.
- Weaver, P. A., K. W. McCleary, L. Lepisto, and L. T. Damonte
 1994 The Relationship of Destination Selection Attributes to Psychological, Behavioral and Demographic Variables. *Journal of Hospitality and Leisure Marketing* 2(2):93-109.
- Woodside, A. G., and S. Lyonski
 1989 A General Model of Traveler Destination Choice. *Journal of Travel Research* 27(4):8-14.
- Zimmer, M. R., and L. L. Golden
 1988 Impressions of Retail Stores: A Content Analysis of Consumer Images. *Journal of Retailing* 64(3):265-293.

Submitted 5 December 1997. Resubmitted 9 September 1998. Accepted 13 October 1998. Final version 2 December 1998. Refereed anonymously. Coordinating Editor: William C. Gartner.